BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: DIK-SSC-X-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 15 | **Paper Code:** DIK-SSC-X-2022 (A)

Q1. Choose the correct option:

Q1. If $(x + 1)(x - 5) = 0$ then the solution are _____

(A) x=(1,−5)
(B) x=(1,5)
(C) x=(−1,−5)
(D) x=(−1,5)

Q2. If a and b are the roots of $4x2 - 3x + 7 = 0$ then the value of $1/a + 1/b$

(A) −3/4
(B) 3/7
(C) −3/7
(D) 4/7

Q3. Direct variation between a and b is expressed as _____

(A) a = b
(B) a = 1/b
(C) a ∝ b
(D) a α 1/b

Q4. If 2a + 1 : 21 :: 4 : 7, then _____

(A) a =13/2
(B) a =11/2
(C) a = 10
(D) a =9/2

Q5. $x^2 + 2/x^2 + 2x + 2$ is_____

(a) Proper fraction
(b) Improper fraction
(c) Rational fraction
(d) None of these

Q6. Which one of the following is an example of disjoint sets?

(a) {0,1,2,3} and {3,2,1,0}
(b) {0,2,4,6} and {2,4,6,8}
(c) {0,3,6,9} and {9,16,25,36}
(d) {0,4,8,12} and {6,10,14,18}

Q7. If R = {(0,0)(8,2),(10,3),(14,12) then Dom R = _____

(a) {0,8,10,14}
(b) {0,2,3,12}
(c) {8,10,14}
(d) {0,10}

Q8. In a set of data 63,65,66,67,69 median is _____

(A) 63
(B) 66
(C) 67
(D) 69

Q9. The formula Σx/n determines:

(A) A.M
(B) Median
(C) Mode
(D) G.M

Q10. Cot θ = _____

(a) sinθ/coseθ
(b) 1/coseθ
(c) Coseθ/sinθ
(d) 1/sinθ

Q11. $\sin^2\theta + \text{cose}^2\theta$ = _____

(a) −1
(b) 1
(c) $2\sin^2\theta\ \text{cose}^2\theta$
(d) None of these

Q12. Pythagoras was an _____

(a) Indian
(b) Greek
(c) Pakistani
(d) Chinese

Q13. In a circle, two chords are equally distance from the centre of the circle. The chords are _____

(a) Congruent
(b) Non-congruent
(c) Parallel
(d) Non-parallel

Q14. The measure of the external angle of a regular hexogen is_____

(A) π/3
(B) π/4
(C) π/6
(D) None

Q15. How many tangents can be drawn from a point outside of circle?

(A) 1
(B) 2
(C) 3
(D) 4

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 36 | **Paper Code:** DIK-SSC-X-2022 (A)

Q2. Attempt any 9 questions of the following:

(i) Solve the equation by completing square: $4x^2 + 12x = 0$

(ii) Evaluate: $(1 + w - w^2)7$

(iii) Use synthetic devision, find the value of k if 2 is a zero of the polynomial $2x^4 + x^3 + kx^2 - 8$

(iv) The sum of two numbers is 11 and the sum of their squares is 65. Find the numbers.

(v) Solve the equation: $(\sqrt{(3x+2)} + \sqrt{x})/(\sqrt{(3x+2)} - \sqrt{x}) = 4/1$

(vi) If 8 person complete work in 10 days then how many clays would 10 persons take to complete the same work?

(vii) Resolve the following fractions into partial fractions: $x^2/(x^2 + 2x + 1)$

(viii) Verify De Morgan's laws for the following sets: U = {1,2,3,.....10}, A = {2,4,6,8,10}, B = {1,3,5,7,9}

(ix) If X = {1,2,3,4} and Y = {5,6,7,8} then write function from X to Y.

(x) A set of data contains the values as 148,145,160, 157,156 and 160, Show that mode > median > mean.

(xi) Find r when $\theta = 3½$ radians and l = 4/7m.

(xii) Prove that $\cos^2\theta - \sin^2 = 2\cos^2\theta - 1$.

SECTION – C

Maximum Marks: 24

Q3. If two chords of a circle are congruent then they will equidistant from the centre.

Q4. If two circle touch externally, the distance between their centres is equal to the sum of their radii.

Q5. Prove that the angle in a semi-circle is a right angle.

Q6. Construct a triangle such that $m\overline{AB} = 3^m$, $m\overline{AC} = 4^n$, and $m\angle A = 60^o$. Draw circum circle of this triangle.